Songyang Gao

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Research Interests

LLM-based Agents, Temporal Forecasting, Auto-Research, AI for Science, Knowledge Graphs.

Education

The Hong Kong University of Science and Technology (Guangzhou) Sep 2025 – Jun 2027 (expected)
Master of Philosophy in Artificial Intelligence *Guangzhou, China*
- Advisor: Prof. Hui Xiong (IEEE, ACM, AAAI, AAAS Fellow)

Nanchang University (211, Double First-Class) Sep 2021 – Jun 2025
Bachelor of Engineering in Artificial Intelligence, Ji Luan Academy *Jiangxi, China*
- Baosteel Scholarship Special Prize, 2024 (**only 15 nationwide**) | Huawei Smart Base Scholarship (ratio: 0.3%)

Publications

Preprints / Under Review
- **S. Gao***, Y. Xia*, S. Liu, H. Xiong. "HypoEvo: Backtest-Reinforced Hypothesis Forecasting over Temporal Citation Graphs." Under review. (**First Author**) [Preprint of an earlier version]
- **S. Gao***, Y. Xia*, H. Xiong. "StreamSQLBench: A Streaming Agent Benchmark for Real-Time Data Pipelines." Under review at VLDB 2027. (**First Author**)
- L. Gong*, T. Wang*, **S. Gao***, Y. Wu, Z. Hao, Y. Xu, L. Ding, H. Xiong. "Agentic Social Strategy Discovery: Learning What Matters from Branched Futures." Under review at ACL Rolling Review (ARR).

Published
- **S. Gao**, W. Xu, X. Wu, J. Liu, B. Wang, T. Wu, W. Wan, Q. Liu. "DMEDH: Diffusion Model-boosted Multiplane Extrapolation for Digital Holographic Reconstruction." **Optics Express**, 2024, 32(18):31920–31938. (CAS Tier 2, JCR Q2, **First Author**) [Paper]
- L. Zhang*, **S. Gao***, M. Tong, Y. Huang, Z. Zhang, W. Wan, Q. Liu. "HoloDiffusion: Sparse Digital Holographic Reconstruction via Diffusion Modeling." **Photonics**, 2024, 11(4):388. (JCR Q3, **Co-first Author**) [Paper]
- Z. Chen, B. Lin, **S. Gao**, W. Wan, Q. Liu. "Imaging through scattering media via generative diffusion model." **Applied Physics Letters**, 2024, 124: 051101. (JCR Q2) [Paper]
- L. Chen, Y. Zhou, **S. Gao**, M. Li, H. Tan, Z. Wan. "ARA-Net: An Attention-Aware Retinal Atrophy Segmentation Network Coping With Fundus Images." **Frontiers in Neuroscience**, 2023, 17: 1174937. (JCR Q2) [Paper]

Patents

- H. Xiong, Y. Xia, **S. Gao**. "Intelligent Encapsulation and Controlled Release System for Big Scientific Data Based on Multi-Stage Workflow and Stream Computing." (2026, Accepted)
- W. Wan, **S. Gao**, B. Wang, W. Xu, Q. Liu. "A Dual-Channel Prior-Constrained Multi-Distance Lensless Digital Holographic Reconstruction Method." ZL 2024107267996. (Student First Author, **Granted**)
- W. Wan, **S. Gao**, Z. Zhang, W. Xu, L. Zhang, Q. Liu. "A Sparse Digital Holographic Image Reconstruction Method." ZL 202410464769.2. (Student First Author, **Granted**)

Research Experience

The Hong Kong University of Science and Technology (Guangzhou) Sep 2025 – Present
MPhil Researcher *Guangzhou, China*
Advisor: Prof. Hui Xiong (IEEE, ACM, AAAI, AAAS Fellow).
- **Deep Research System for Big Scientific Data** (From National Key Research and Development Program of China). Researching and designing Agentic Systems for Large-Scale Data Tasks. As first author, built StreamSQLBench, a benchmark of 402 streaming SQL tasks over 51 databases (29.4M rows, 7 domains). Each query runs on a containerized Kafka and Flink harness that replays events in event-time order. Proposed StreamSQLAgent, a multi-agent framework that reaches 79.60% accuracy, compared with 68.16% for the strongest baseline Spider-Agent, while using one-sixth of its tokens. Outcomes include a paper under review at VLDB 2027, 1 filed patent, SciFlow and DeepScience.
- **Backtest-Reinforced Hypothesis Forecasting (HypoEvo).** Treated research hypothesis generation as a forecasting task over temporal citation graphs. Built an automated pipeline that collects time-consistent snapshots of 61,671 accepted papers from the scholarly Web. Trained a 7B LLM with GRPO, rewarding each hypothesis by how well later accepted papers realize it. The model raises Hit@10 from 22% to 46% and outperforms six baselines, five of which run on the much larger GLM-5.3. It also transfers to Earth science, chemistry and biology without retraining. This work extends Graphs of Research (GoR). [Preprint]

- **Autonomous Research Agents (Redigg).** Built Redigg, an autonomous research-agent system for scientific discovery that combines a tool-calling loop, a modular skill registry, memory storage, and a React dashboard for streaming reasoning and tool traces. Implemented core research skills for paper search, PDF reading, literature synthesis, gap analysis, and report export, with HTTP/SSE and A2A interfaces for integration. [Code]

LIVE – Laboratory of Imaging and Vision Expression, Nanchang University Oct 2022 – Jun 2025
Research Assistant *Jiangxi, China*
Advisors: Prof. Qiegen Liu and Wenbo Wan

- Trained diffusion generative models as learned priors for inverse problems in computational imaging, including digital holography and imaging through scattering media. The methods improve reconstruction quality while reducing the cost of optical hardware.
- Mainly responsible for technical solution design, code implementation, simulations, real-world experiments, and manuscript writing. Developed diffusion-prior methods for digital holographic reconstruction, resulting in 3 journal papers, 2 granted patents, and a best-poster award.

Beijing Aishi Future Technology Co., Ltd. Mar 2025 – Aug 2025
AI Algorithm Research Intern *Beijing, China*

- Researched pose-driven human video generation. Retargeted skeleton sequences across identities and conditioned video diffusion models on reference images and temporal pose sequences.
- Conducted data processing, model training, and quantitative evaluation across multiple baselines including RealisDance, Animate-X, and MTVCrafter.

Institute for AI Industry Research (AIR), Tsinghua University Jan 2024 – Mar 2024
Research Intern (Winter Camp) *Beijing, China*
Supervisor: Yan Wang

- Used Stable Diffusion for text-guided image editing to generate diverse weather scenes, which served as training data for autonomous driving models.
- Reproduced the open-source Stable Diffusion project and adapted it to weather scene editing.

Professional Experience

Singtel PIE Program Dec 7–14, 2025
Group Leader *Singapore*

- Designed and proposed **SmartShield**, a multi-modal scam detection and prevention system via rule-LLM synergy, combining field research in Singapore with AI-driven solutions for real-world fraud prevention.

Tianjin Zhishan Information Technology Co., Ltd. Dec 2022 – Sep 2025
Technical Lead *Tianjin, China*

- Led the technical side of a college student growth community with 700+ paying users. Built and deployed RAG-based LLM applications and domain-specific agent Q&A systems in production (junyisocial.com), as well as the brand website (zhishanzhe.com).

Honors & Awards

- China International College Student Innovation Competition, National Silver Award Oct 2024
- The Best Poster Award of the 6th Symposium on Computational Imaging Technology and Applications May 2024
- Outstanding Academic Correspondent in CAAI Transactions on Intelligence Technology (IF:7.985) May 2024
- The 14th "Challenge Cup" National Entrepreneurship Competition, National Bronze Award Nov 2024
- The 1st prize in the 5th Global Campus Artificial Intelligence Algorithm Elite Competition Dec 2023
- The 1st Prize in the 14th Blue Bridge Cup Special Competition Nov 2023
- The H Prize (2nd Prize) in the USA Mathematical Contest in Modeling Jul 2023
- The 3rd Prize in the 25th China Robot and Artificial Intelligence Competition May 2023
- The 1st Prize in the 13th MathorCup College Mathematical Modeling May 2023
- The 1st Prize in the Big Data Competition at the 3rd MathorCup College Mathematical Modeling Mar 2023

Selected Open-Source Projects

- **ds-spec-loop.** Open-sourced a reusable Spec-programming workflow that structures requirement clarification, architecture decisions, specification, task decomposition, verification, and convergence across code, tests, and documentation. [Code]

Skills

- **Programming & Data Systems:** Python, PyTorch, C/C++, SQL, Apache Flink, FastAPI, Next.js
- **AI/ML:** LLM-based Agents, LLM Fine-tuning, Diffusion Models, Computer Vision
- **Tools:** Docker, Git, Linux, vLLM, Slurm, Cloud Deployment